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US Real Estate Sector Overview & Trends

June 2023

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Executive Summary

Introduction

- This presentation a) offers an overview of the Real Estate ("RE") industry and b) presents trends in the RE and property markets, more generally
- RE as a whole in the United States, inclusive of currently operating commercial buildings and commercial developments, contributed $2.3 trillion to US GDP in 2022 (>9% of total), generated nearly $832 bn in personal earnings and supported 15.1 mm jobs in the United States

Real Estate is Facing Significant Pressure

- RE, and its derivative sectors, has been directly impacted by the Covid-19 pandemic, and many of the themes that are present in the market today are directly linked to policies and changes that began (or were otherwise accelerated) in early 2020, including:
  - Business travel evolution
  - Work from home dynamics
  - Growth of e-commerce relative to in-store shopping
  - Inflation, and the resulting rise in interest rates
  - Housing affordability
  - Shifting demographics (age and geography)
- RE values are falling and, in particular, office and retail assets are facing significant pressure as a result of the current macroeconomic environment
  - Rent growth has been anemic as landlords seek to attract tenants and, especially for office properties, vacancy and availability are at levels experienced during the Global Financial Crisis
  - As a result, RE owners and lenders are increasingly concerned about the ability of assets to cover rising debt service costs in an inflationary environment and to ultimately refinance debt as it matures

Restructuring Opportunity

- Total commercial RE mortgage debt amounts to ~$4.5 tn and debt attributable to office and retail, alone, is ~$2.5 tn. In addition, there is an incremental ~$722 bn of corporate debt issued by real estate companies, per Bloomberg
- <u>To give a sense of the restructuring opportunity, a recent JPM forecast estimated that the value loss on office and retail RE debt is likely going to be greater than $120 bn in the current market</u>
- In addition, from a corporate perspective, there are multiple situations that may present opportunities to evaluate liability management exercises
- We have developed, and can share with parties if desired, an internal screen of potential distressed opportunities in the RE industry that we are presently monitoring

Ducera's Real Estate Experience

- Ducera has extensive experience advising on complex situations across the Real Estate industry:

BridgeStreet Worldwide

Advisor to Secured Creditors in connection with BridgeStreet's sale to Sorrento Asset Mgmt.

CBL & Associates

Advisor to Wells Fargo in CBL & Associates' Chapter 11 restructuring

Redfin

Advisor to Redfin in connection with its Proposed $608mm Acquisition of RentPath

WeWork

$3.9 billion Consensual Liability Management and New Money Transaction

Toys-R-US

Advisor to independent board of directors in connection with Toys "R" Us's Chapter 11 Restructuring

Azabu Buildings Co.

Advisor to the Committee of Unsecured Creditors in connection with Azabu's Chapter 11 restructuring

HCR ManorCare

Advisor to First Lien Lenders in connection with out-of-court restructuring

Simon Baron Development

Advisor to Simon Baron in connection with the recapitalization of certain New York City assets

Dubai World

Advisor to Dubai World in connection with the restructuring of the Las Vegas City Center JV

Ventas

Advisor to company on capital structure matters

FDIC

Investor in multiple loan pools during Global Financial Crisis

666 Fifth Avenue

Investor in 666 Fifth Avenue's Single Asset CMBS Restructuring

Colony Financial

Internal advisor to company in development of mortgage REIT's business model and IPO

Extra Space Storage

Advisor to company in connection with various M&A and financing transactions

GICSA

Advisor to company in connection with capital structure matters

Represents select transactions, inclusive of those completed at prior firms

I. Real Estate Landscape

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Overview of US Real Estate Industry

REITs

Office
- Office buildings leased out to companies across different sectors

Gaming
- Casino assets and other large-scale gaming venues that are leased to operators

Retail
- Shopping centers and other consumer facing sales-based property types leased to retail tenants

Mortgage
- Finance companies that originate and invest in individual mortgages and mortgage-backed securities

Multi-Family
- Apartment buildings and other residential properties containing multiple units on a given plot of land leased to tenants

Single Family
- Rental homes built for singular tenants as part of larger communities

Healthcare
- HC-related real estate incl. senior living communities, hospitals, medical offices, outpatient facilities, life science innovation and research properties etc., leased to operators

Hotel
- Hotel assets and portfolios in which a third party operates the property, often under a specific brand

Data Centers
- Facilities to store data and leased to tech companies and generally provide products and services to keep servers and data safe (e.g. uninterruptable power, air-cooled chillers, etc.)

Industrial
- Properties provided for commercial use related to logistics, manufacturing, and warehousing

Infrastructure
- Infrastructure real estate including fiber cables, wireless infrastructure, telecommunications towers and energy pipelines leased to telecommunications / other operators

Other / Specialty
- Portfolios of niche property types
- These may include self storage, manufactured housing, amusement parks, prisons, etc.

Non-REITs

Land Developers
- Acquisition and development of various property types into commercial or residential buildings or homes

Residential Brokerage
- Medium for buyers and sellers to connect and enable them to complete transactions

Information Services
- Providers of real estate focused data and industry reports

Hospitality Brands
- Management companies that enable short-term rental of stays / experiences for consumers and businesses

Banks / Non-Bank Lenders
- Commercial and residential real estate lenders / underwriters

Other / Specialty
- Companies engaging in unique / niche real estate applications (i.e., proptech)

Overview of US Real Estate Industry (cont.)

REITs

Office
brandywine; Cousins; COPT; Corporate Office Properties Trust; Piedmont Office Realty Trust; Highwoods; SL Green Realty Corp; Vornado Realty Trust; Easterly Government Properties Inc; Boston Properties

Gaming
Gaming & Leisure Properties; Penn National; VICI; MGM Resorts International

Retail
Brixmor; Macerich; Simon Property Group; Regency Centers; Kimco Realty; Acadia; Whitestone REIT; SITE

Mortgage
Blackstone Real Estate Income Trust; New Residential Investment Corp.; Starwood; AGNC Investment Corp

Multi-Family
Equity Residential; veris Residential; elme; American Homes 4 Rent; AvalonBay Communities, Inc; NexPoint Diversified Real Estate

Single Family
invitation homes; Colony Starwood Homes; Tricon; American Homes 4 Rent; NexPoint Diversified Real Estate

Healthcare
Community Healthcare Trust; Physicians Realty Trust; Sabra Health Care REIT, Inc.; MPT Medical Properties Trust; Omega; NHI National Health Investors

Hotel
RLJ Lodging Trust; Ryman Hospitality Properties, Inc.; Park Hotels & Resorts; pebblebrook hotel trust; HR REIT

Data Centers
Digital Realty; QTS; Equinix; CyrusOne Built for Tomorrow. Ready Today.

Industrial
EastGroup; americold; Prologis; First Industrial Realty Trust; Plymouth REIT

Infrastructure
Uniti The Communications REIT; Crown Castle; American Tower; CorEnergy; SBA

Other / Specialty
CubeSmart; GEO; Sun Communities, Inc.; Equity LifeStyle Properties

Non-REITs

Land Developers
Five Point; United Construction & Development Group; Extell; Brookfield

Residential Brokerage
Coldwell Banker; HomeServices of America A Berkshire Hathaway Affiliate; kw; Compass

Information Services
Avison Young; CoStar Group; CBRE; REAL Capital Analytics

Hospitality Brands
Holiday Inn; Hyatt; Marriott Hotels · Resorts · Suites; Four Seasons Hotels and Resorts

Banks / Non-Bank Lenders
Bank OZK; M&T Bank; Valley National Bank; Huntington

Other / Specialty
Zillow; Apartments.com; Buildium Property Managed.; StreetEasy

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II. Themes and Trends

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Key Themes Impacting Real Estate

	Description
1 Business Travel Evolution	▪ Reduction in corporate business travel versus pre-Covid ▪ ~20% of corporate travel expected to go virtual, per Morgan Stanley
2 Work From Home vs. Return to Office	▪ According to PWC, only 49% of workers expect to come into the office every day versus 66% pre-Covid ▪ Vacancy and availability rates in the US are near levels seen during Global Financial Crisis
3 In-Store vs. Online Shopping	▪ Rise in e-commerce began before the Covid pandemic and accelerated due to the pandemic ▪ E-commerce as a percentage of total sales was 14.7% as of Q4 2022, up from 11.1% in Q4 2019, per the Federal Reserve
4 Inflation	▪ Inflation continues to remain above the Federal Reserve's stated goal of 2% (core PCE of 4.9% YoY in Mar '23) ▪ Higher prices divert cash flow from businesses and consumers, leaving less cash available to pay rent ▪ Increases in costs associated with property management that have not been passed onto tenants have pressured margins for RE entities
5 Interest Rates	▪ Federal Reserve has increased the Fed Funds Rate from zero in early 2022 to a range of 5.00% to 5.25% at present, and may go higher to curb inflation ▪ Increasing interest rates make it more expensive for real estate owners to refinance their debt ▪ Furthermore, higher interest rates have resulted in higher cap rates, which has led to value declines across the real estate industry
6 Housing Demand & Supply	▪ Housing affordability remains challenged and rising mortgage rates puts pressures on buyers ▪ New / existing home sales are in decline; pending home sales set new record lows in the final months of 2022 and the early months of 2023 ▪ Existing and new inventory growth has slowed and months of supply registered at 3.3 months in February '23, significantly below long-term averages
7 Demographic Shifts	▪ US population growth recovered in 2022, increasing by 0.4% YoY, driven primarily by net international migration; for comparison, in 2012, population growth was 0.9% YoY ▪ Pandemic has resulted in meaningful population losses in certain cities, including San Francisco (down 6.3% YoY in 2021), New York City (down 3.5%), Washington DC (down 2.9%) and Boston (down 2.9%) ▪ These changes have widespread impacts on office, apartment, and retail REITs
8 Tech Sector Layoffs	▪ Technology firms have laid off more than 330K employees between 2022 and 2023, creating an impact on demand for real estate across many sub-sectors

Key Sectors Under Distress in Commercial Real Estate

	Overview	1	2	3	4	5	6	7	8
Office	▪ Headwinds from remote / hybrid work persist ▪ For several years, the tech and life science sectors were the two main areas driving demand for office space ▪ In 2H22, tech sector began to pull back notably with RIFs and space give-backs— a trend that has continued into 2023	✓	✓		✓	✓		✓	✓
Retail	▪ Increases in digitally influenced sales have resulted in headwinds for brick-and-mortar retailers ▪ Though decreases in new supply should help realign supply / demand in the sector, high inflation, rising interest rates and labor shortages remain challenging			✓	✓	✓			✓
Data Center	▪ Increases in labor and power costs unable to be passed onto customers ▪ Development, a core earnings driver, subject to headwinds due to supply chain pressures and yield compression ▪ Demand from enterprise customers remains strong though hyperscale is a concern as large-scale tech spending pulls back				✓	✓			✓
Hotel / Lodging	▪ Overall hotel demand is still down vs. pre-pandemic levels ▪ Higher wages and operating costs outpace top line growth resulting in margin pressure for operators ▪ Headwinds relating to declines in corporate travel (corporate travel historically closely correlated to earnings growth)	✓	✓		✓	✓			✓
Multi-Family	▪ Vacancy rates continue to rise with additional 2.6% supply entering the market in 2022 ▪ Tailwinds relating to decrease in the affordability of owning vs. renting		✓		✓	✓	✓	✓	
Single-Family	▪ New homes sales down 17% YoY in February 2023 and purchase applications down 38% YoY in March ▪ Rising mortgage rates pose a threat to growth		✓		✓	✓	✓	✓	
Healthcare	▪ Certain operators sustained by government relief during Covid, which "masked" underlying weakness ▪ Tenants facing labor shortages in certain sub-sectors (especially hospitals and skilled nursing), which is hurting performance ▪ Medicare/Medicaid reimbursement poses consistent issue				✓	✓		✓	

1 Business Travel Evolution
2 WFH vs. Return to Office
3 In-Store vs. Online Shopping
4 Inflation
5 Interest Rates
6 Housing Demand and Supply
7 Demographic Shifts
8 Tech Sector Layoffs

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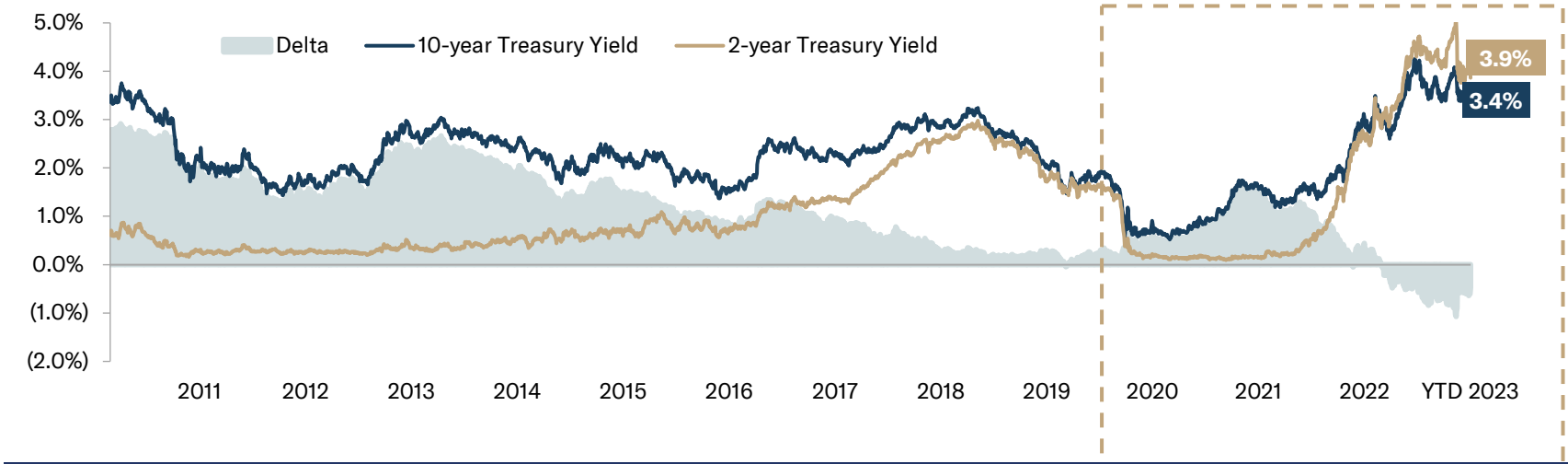
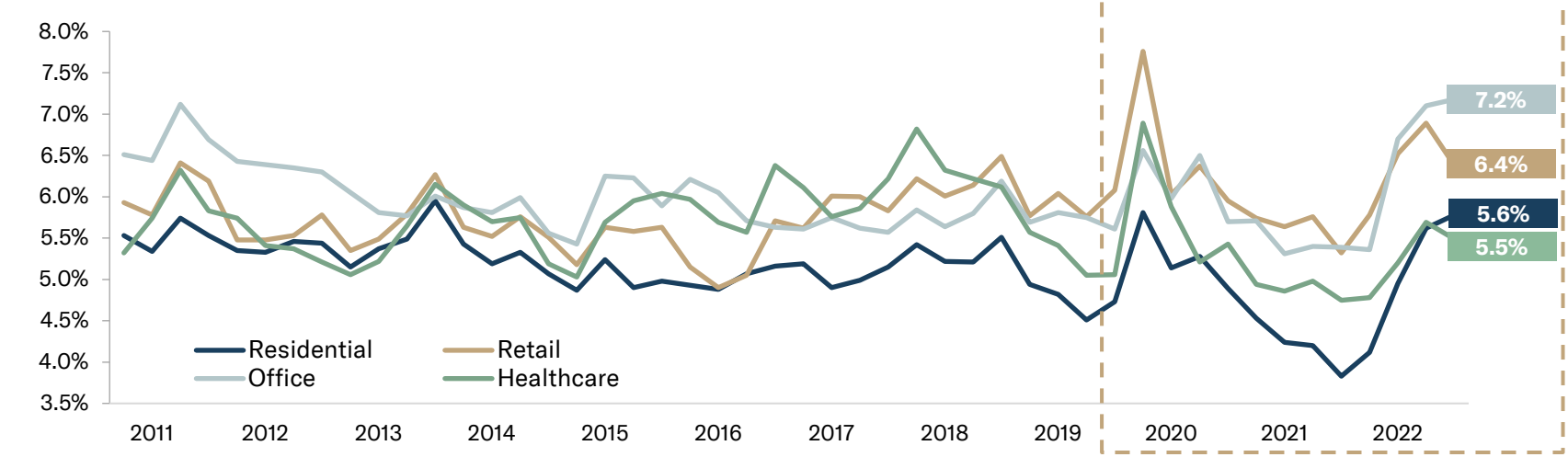
Rising Interest Rates Have Resulted in Higher Cap Rates…

Treasury Yield



Cap Rate by Property Type[1]



Sources: Bloomberg, Refinitv

(1) Aggregate cap rate data for North America US Equity REITs by sector

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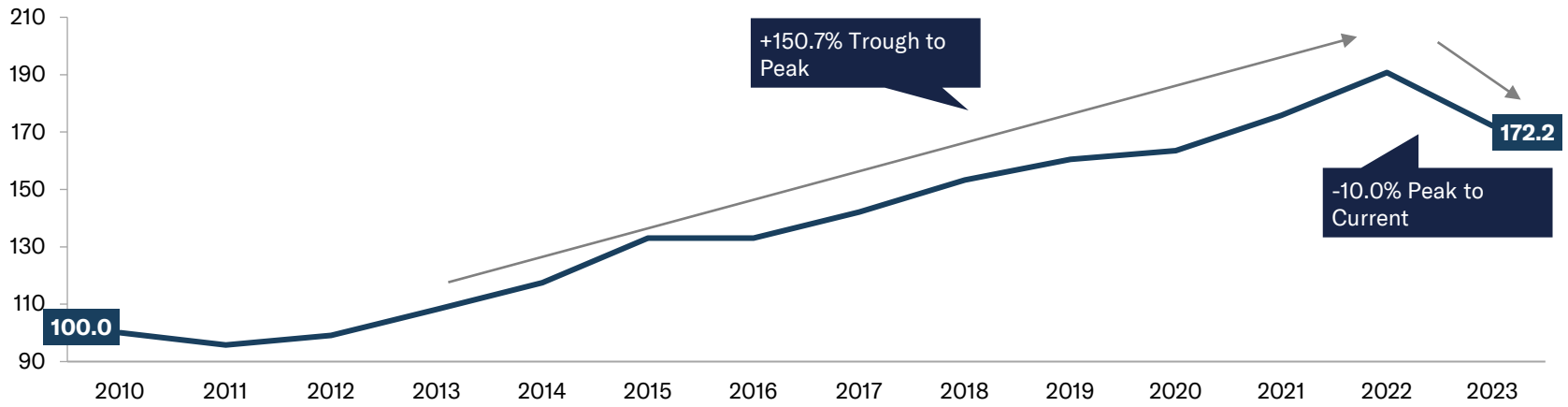
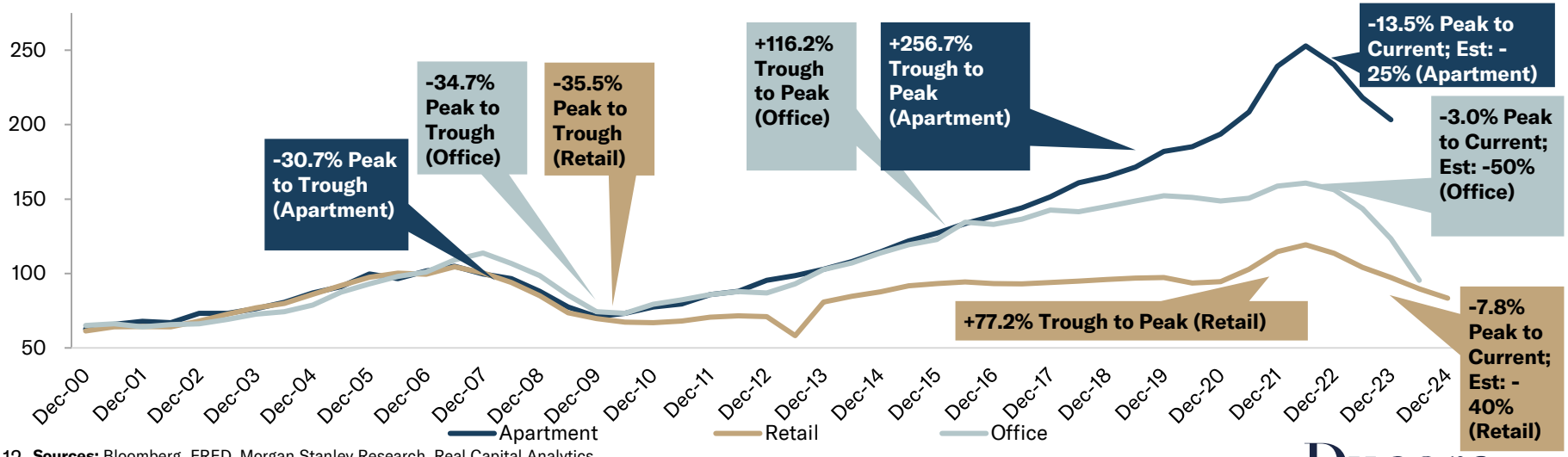
...Which is Reducing Asset Valuations

Commercial Real Estate Prices (Indexed)



Commercial Real Estate Prices by Sector



 Sources: Bloomberg, FRED, Morgan Stanley Research, Real Capital Analytics

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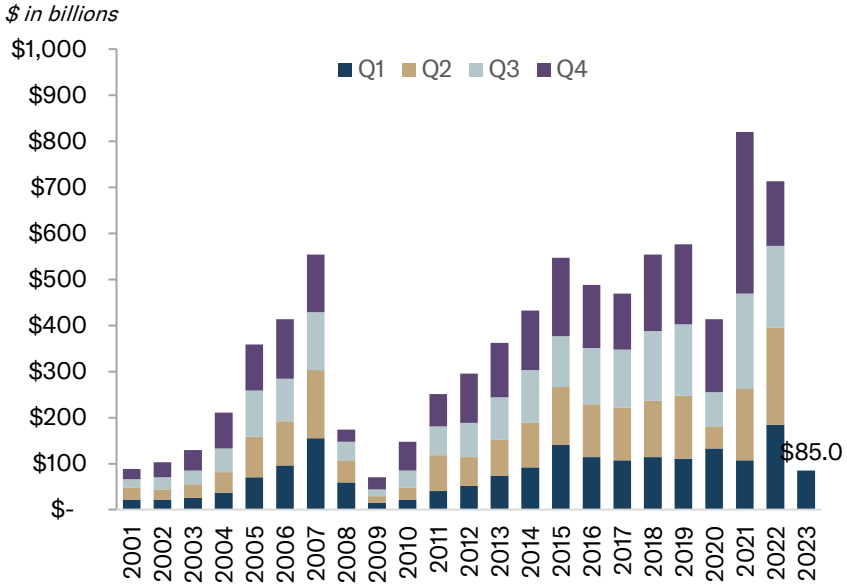
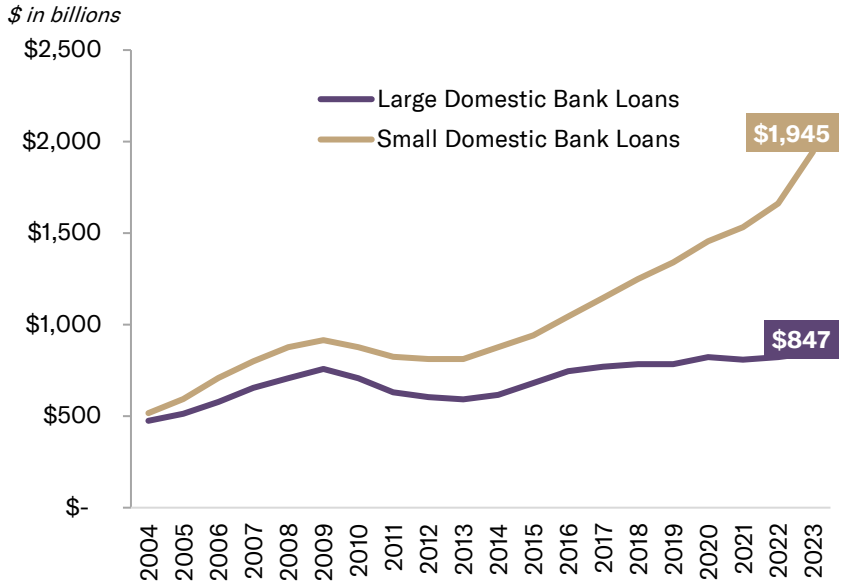
Transaction Volumes Declining Given Macroeconomic Environment

- Commercial RE transaction volume of $85 bn in 1Q23 is down 56% YoY and is 31% below the 10-year average in Q1 of $122.6 bn
  - Office down 78% YoY
  - Apartments down 64% YoY
  - Retail down 64% YoY
  - Industrial down 54% YoY

Commercial RE Transaction Volumes Are Declining...



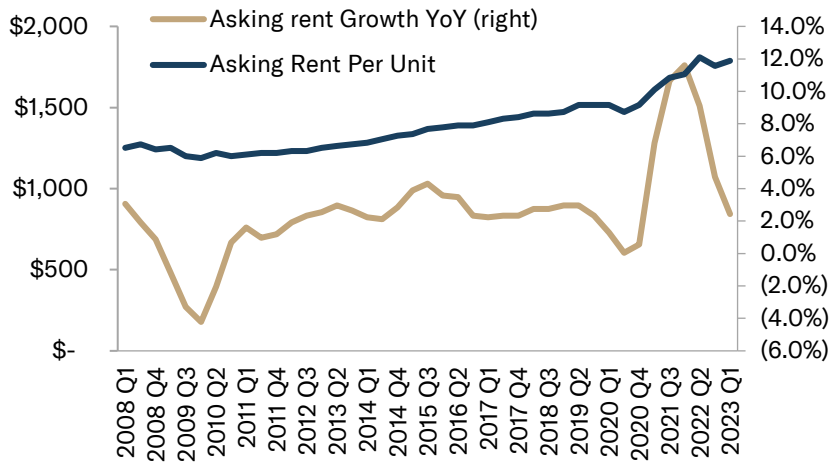
...And Loans Outstanding Will be Under Pressure



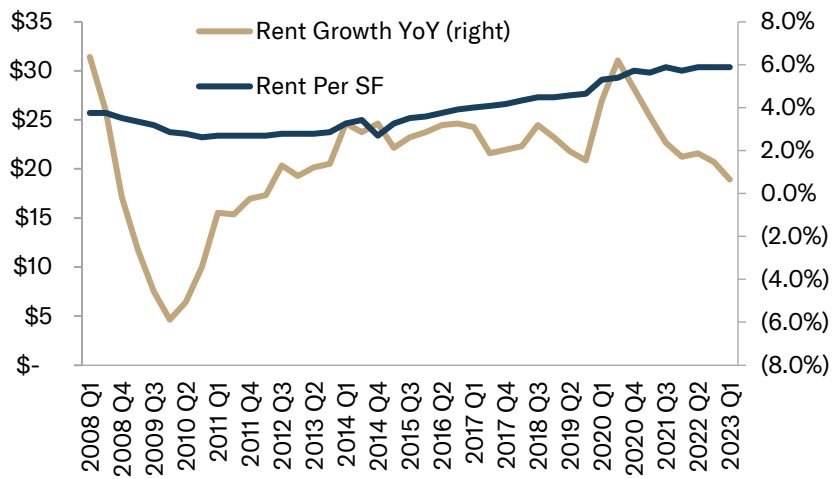
 Sources: Bloomberg, FRED, Morgan Stanley Research, Real Capital Analytics, CoStar, JPMorgan

Property Owners Have Slowed Rent Growth to Attract Tenants

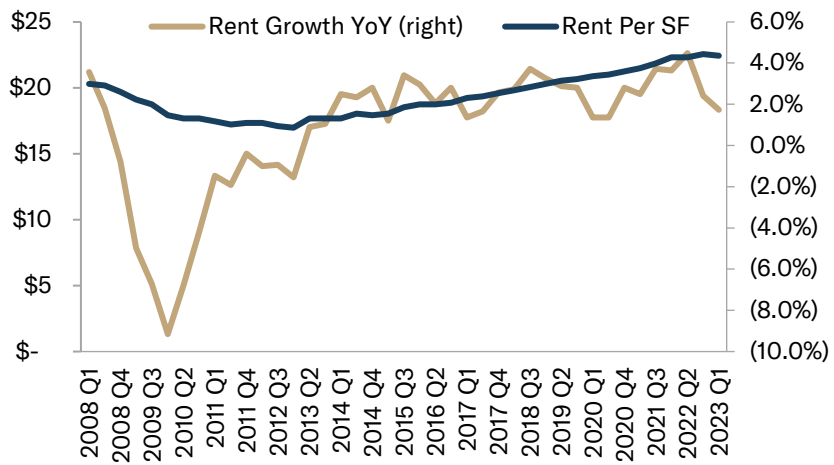
Multi-Family



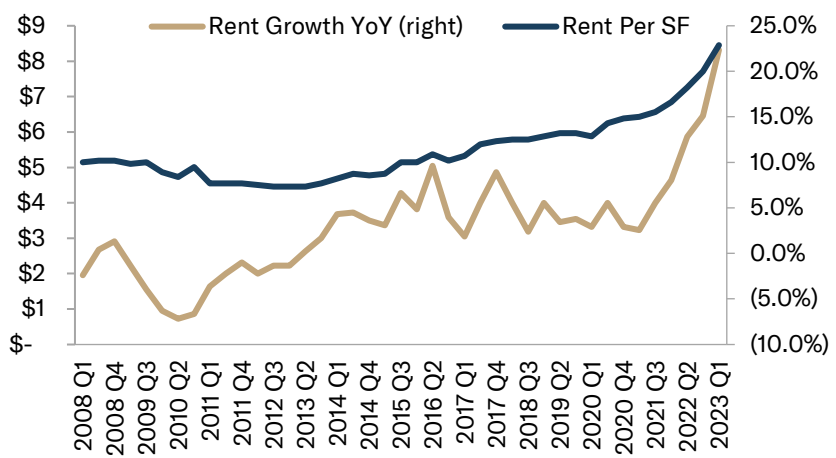
Office



Retail



Industrial

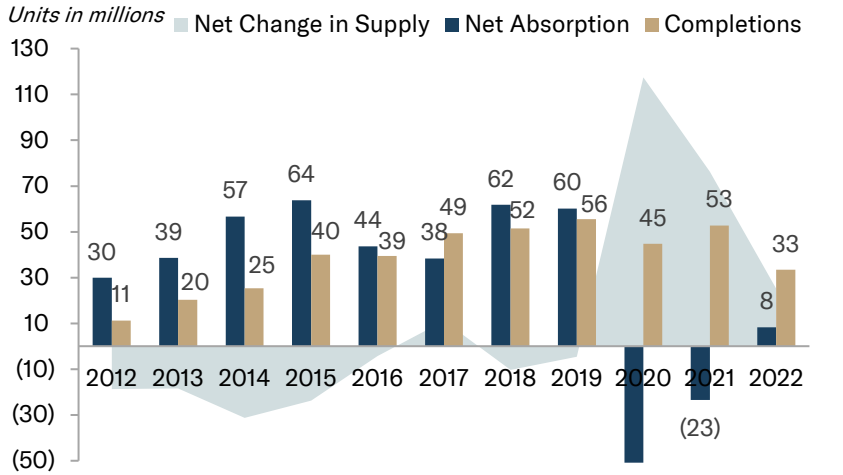


 Sources: Bloomberg, FRED, Morgan Stanley Research, Real Capital Analytics, CoStar, JPMorgan

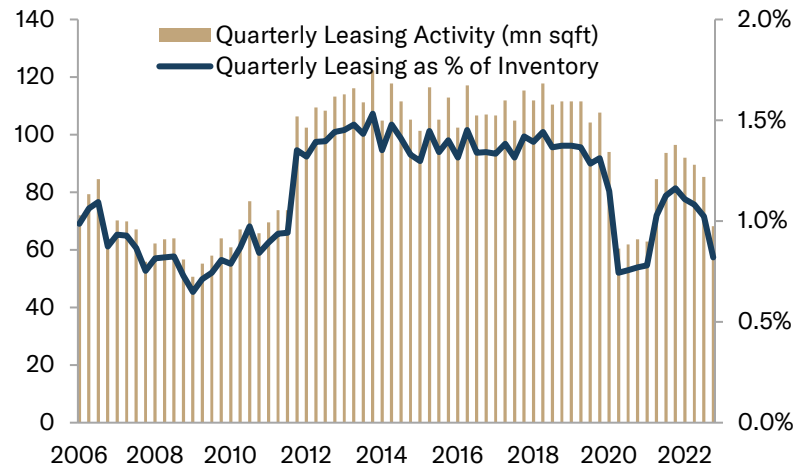
Office Market Fundamentals Are Weak

- Office market continues to show weakness, as evidenced by rising office supply, declining leasing volumes, GFC-era vacancy rates and increased rent discounts for subleases

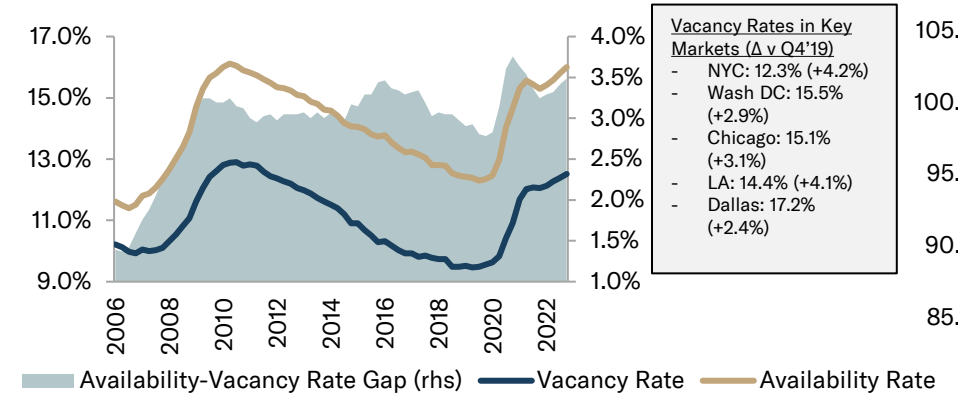
Office Supply and Demand



Leasing (mm SF) and % Inventory Near Pandemic Levels

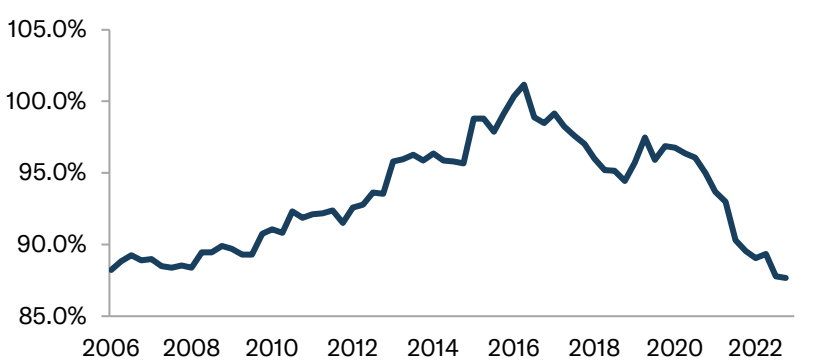


National Vacancy and Availability Rates Near GFC Levels



Vacancy Rates in Key Markets (Δ v Q4'19)
- NYC: 12.3% (+4.2%)
- Wash DC: 15.5% (+2.9%)
- Chicago: 15.1% (+3.1%)
- LA: 14.4% (+4.1%)
- Dallas: 17.2% (+2.4%)

Sublease Rent Discounts Show Continued Weakness



Sources: Bloomberg, FRED, Morgan Stanley Research, Real Capital Analytics, CoStar, JPMorgan

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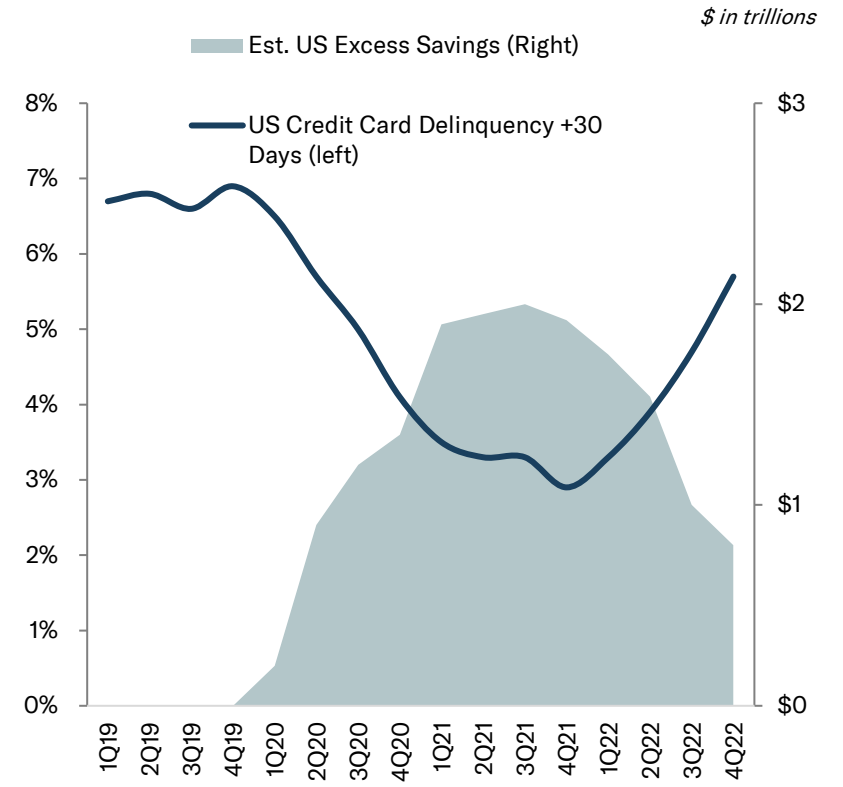
Headwinds Facing Retail Real Estate

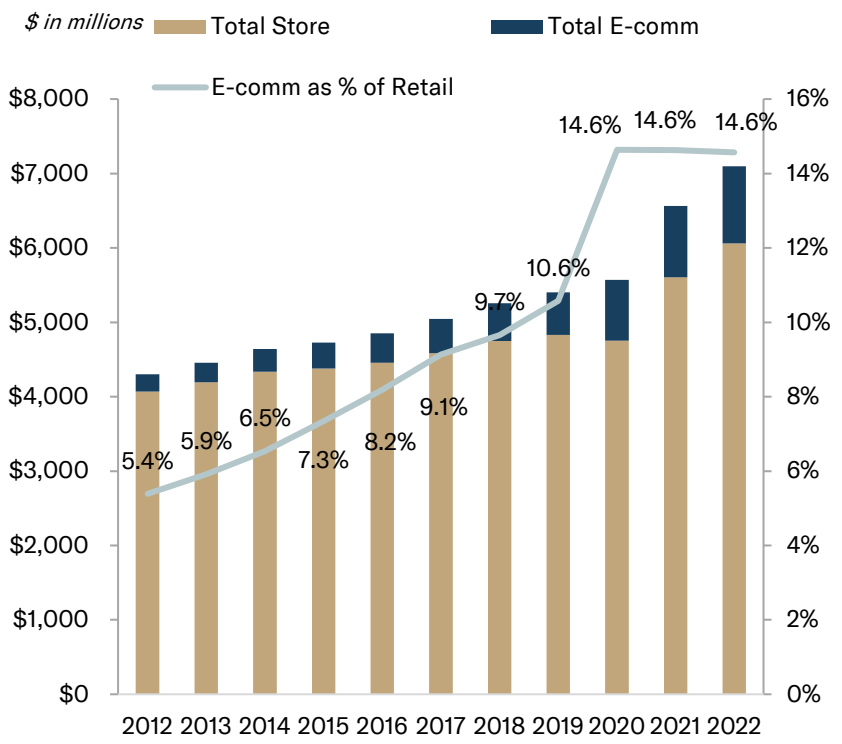
- Consumer savings will likely soon deplete and credit card delinquencies are rising, which poses a threat to retail sales. In addition, there is the constant growth of e-commerce, which serves as an underlying risk to brick and mortar

Savings and Credit Card Delinquencies



Excess savings represent an estimate of additional savings consumers stockpiled due to changes in government stimulus and spending patterns since the onset of covid-19

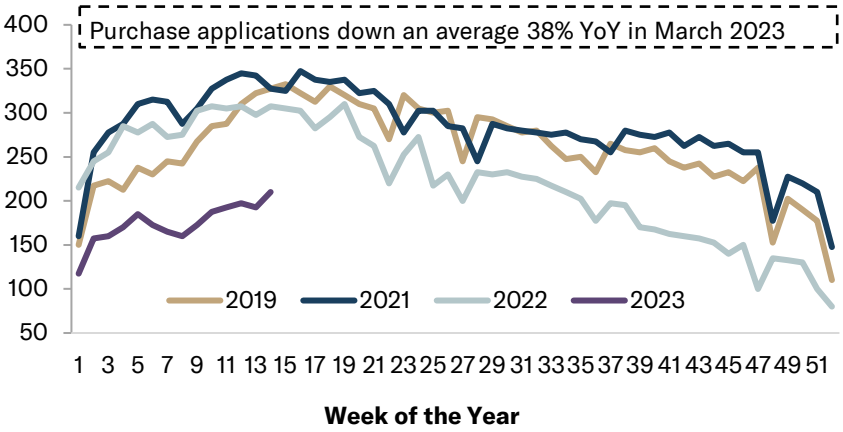
E-Commerce as % of Retail Sales



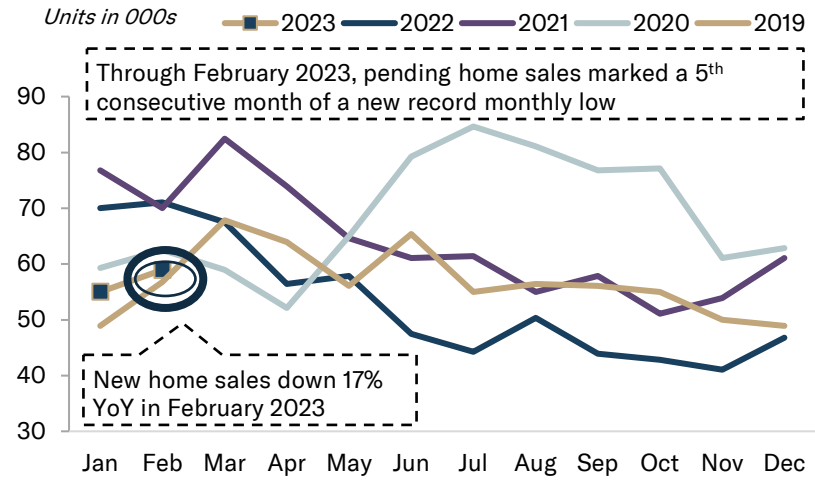
 Sources: Bloomberg, FRED, Morgan Stanley Research, Real Capital Analytics, CoStar, JPMorgan, Green Street

Housing Market Also Facing Pressure Due to Rising Mortgage Rates

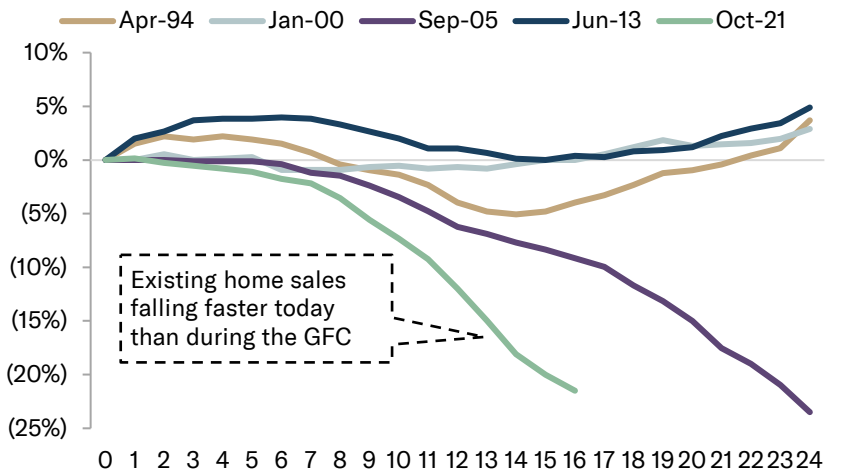
Purchase Applications



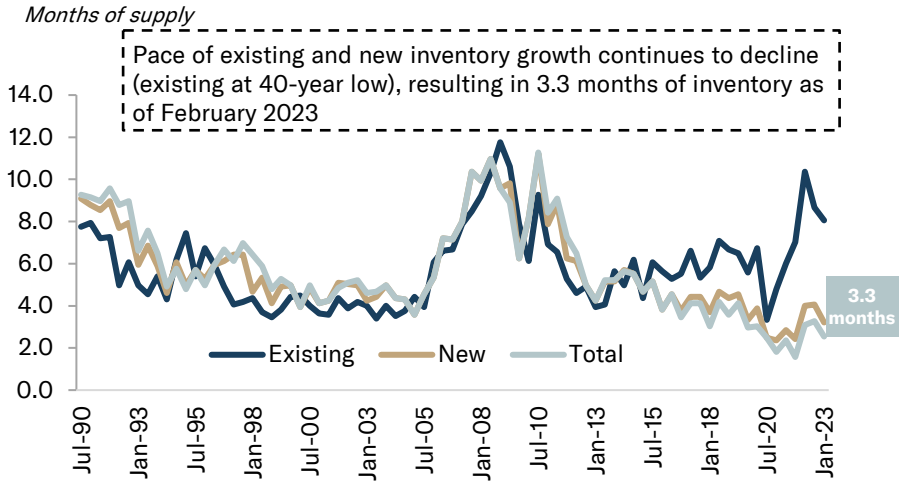
New Home Sales



Existing Home Sales



Housing Inventory



 Sources: Bloomberg, FRED, Morgan Stanley Research, Real Capital Analytics, CoStar, JPMorgan

Commercial Real Estate Financing Landscape

- According to JPM, total commercial RE mortgage debt outstanding is about $4.5 trillion, of which 38% is with US banks; excluding multi-family exposure, total commercial RE debt is $2.5 trillion, of which 44% is with US banks
  - US commercial banks also held $455 bn of construction and land development loans as of early March 2022
  - $448 bn of maturing RE loans in 2023, of which ~60% are held by banks
- Life insurers are major investors in commercial mortgages, both in whole loans ($655 bn) and CMBS ($199 bn)
- CMBS accounts for 22% or $549 bn of commercial RE debt (excluding multi-family), of which $185 bn is exposed to office
- JPM forecasts an 8.6% loss rate for office exposure, which implies $82 bn of losses, and 6.4% loss rate for retail, which implies $39 bn of losses
  - Of these losses, 46% expected to occur in banking industry ($55 bn), 20% in private label CMBS ($24 bn), 19% for life insurers ($23 bn) and 7% among REITs ($9 bn); loss estimates are high level and may not account for relative LTVs attributable to each lender type

($ in billions)	Commercial and Multi-family	% of CRE	Multi-fam only	CRE x Multi-fam	% CRE x Multi-fam	Est Office (39% of market)	Est Retail (24% of market)	% of Office and Retail	% CRE x Multi-fam Losses	Office Losses	Retail Losses
Bank and Thrift	1,711	38%	584	1,127	44%	444	270	46%	68	38	17
Angency and GSE	926	21%	926	0	-	0	0	-	0	0	0
Life Insurance Companies	655	15%	189	466	18%	184	112	19%	28	16	7
Private Label CMBS	663	15%	114	549	22%	185	124	20%	34	16	8
REITs	176	4%	7	169	7%	67	40	7%	10	6	3
State and Local Government	133	3%	111	22	1%	9	5	1%	1	1	0
Federal Government	91	2%	11	80	3%	32	19	3%	5	3	1
Nonfarm, Noncorporate Business	38	1%	21	17	1%	7	4	1%	1	1	0
Other Insurance	31	1%	0	31	1%	12	7	1%	2	1	0
Finance Companies	30	1%	5	25	1%	10	6	1%	1	1	0
Private Pension Funds	25	1%	0	25	1%	10	5	1%	1	1	0
Nonfinancial Corporate Business	23	1%	1	22	1%	9	5	1%	1	1	0
State and Local Retirement Funds	5	-	2	2	-	1	1	-	0	0	0
Household Sector	1	-	0	1	-	1	0	-	0	0	0
Total	**4,507**		**1,972**	**2,535**		**968**	**600**		**154**	**82**	**39**

 Sources: Bloomberg, FRED, Morgan Stanley Research, Real Capital Analytics, CoStar, JPMorgan

Analyzing Bank Exposure to Commercial Real Estate Debt

- Certain regional banks have commercial RE debt exposures that exceed their common equity tier 1 ratios

Top 25 US Banks with Commercial Real Estate Exposure and % of Capital

	Ticker	Adj. US CRE	US Multi Family	Adj. CRE exc. Multi Family	% CET1 Capital
	WFC	100,941	26,684	74,257	57%
	BAC	40,933	5,774	35,159	20%
	JPM	106,427	79,276	27,151	13%
M&T Bank	MTB	27,028	5,819	21,210	136%
	TFC	25,633	4,630	21,003	55%
	PNC	29,247	10,852	18,395	47%
	COF	27,444	10,790	16,654	38%
	USB	21,632	6,615	15,017	34%
	CFG	22,213	8,441	13,772	75%
	TD	19,148	5,487	13,661	32%
	MS	15,693	2,087	13,606	20%
Valley National	VLY	20,597	7,646	12,951	310%
Signature Bank	SBNY	30,378	18,931	11,447	126%
East West Bank	EWBC	15,341	4,118	11,223	186%
	C	19,150	8,702	10,448	7%
	FRC	30,001	20,364	9,637	71%
	RBC	12,450	2,950	9,500	58%
	HBAN	12,429	3,216	9,213	72%
Synovus	SNV	11,020	1,870	9,150	192%
Webster Bank	WBS	15,316	6,232	9,085	161%
CIBC U.S.	CIBC	13,299	4,569	8,730	119%
	FCNCA	10,916	2,417	8,499	93%
Western Alliance	WAL	8,586	359	8,227	172%
First Horizon	FHN	10,012	1,976	8,036	118%
	GS	8,730	805	7,925	8%
	Other BHCs	676,709	236,551	440,158	-
	Total	**1,331,274**	**487,160**	**844,114**	
	MBA CRE	1,710,987	584,144	1,126,843	
	Capture %	**78%**	**83%**	**75%**	

Commercial Real Estate & Construction Loan Exposure for US Banks with >$30 BN Assets

Ticker	% Domestic Loans	% CET1 Capital	Ticker	% Domestic	% CET1 Capital
VLY	37%	396%	CBSH	19%	90%
SNV	28%	250%	CMA	13%	86%
UMPQ	23%	206%	TFC	10%	79%
EWBC	26%	195%	WFC	11%	74%
MTB	23%	192%	RF	8%	65%
WBS	22%	187%	PNC	8%	64%
ZION	21%	179%	KEY	8%	63%
ASB	18%	174%	FITB	7%	58%
FHN	19%	163%	USB	7%	57%
BKU	19%	163%	COF	6%	45%
CFR	34%	161%	NTRS	8%	35%
HWC	21%	148%	BAC	5%	27%
SBNY	17%	143%	BK	6%	23%
WTFC	15%	136%	MS	6%	20%
FCNCA	17%	129%	SIVB	4%	19%
BOKF	22%	109%	JPM	3%	17%
CFG	13%	105%	STT	6%	12%
BPOP	21%	98%	C	4%	12%
FRC	8%	97%	GS	6%	12%
HBAN	10%	92%	ALLY	1%	10%

 Sources: Bloomberg, FRED, Morgan Stanley Research, Real Capital Analytics, CoStar, JPMorgan

Analyzing Bank Exposure to Office/Retail Commercial Real Estate Debt

- Certain regional banks have significant exposure to office and retail commercial RE debt

Select US Banks with Disclosed Exposure to Office and Retail Commercial RE debt

	($ in millions)	CRE (x owner-occupied, multi-fam, C&D)	Office	Retail	Total	% Adj. CRE	CRE (x owner-occupied, multi-fam)	C&D	30% C&D (Retail & Office Est)	Office, Retail, C&D % CET1 Capital	CET1 Capital
Cullen/Frost Bankers	CFR	3,657	1,833	917	2,750	75%	5,747	2,014	604	89%	3,751
	WFC	73,339	32,478	21,287	53,765	73%	96,629	23,573	7,072	46%	133,527
	CFG	13,741	6,300	2,200	9,600	70%	19,858	4,618	1,386	59%	18,575
	BAC	36,076	18,230	5,830	24,060	67%	48,022	10,465	3,139	15%	180,060
Wintrust Bank	WTFC	4,085	1,393	1,294	2,687	66%	5,693	1,487	446	75%	4,162
	PNC	18,732	9,123	2,855	11,978	64%	25,638	6,400	1,920	35%	39,685
BankUnited	BKU	4,448	1,875	870	2,745	62%	4,728	294	88	101%	2,807
	USB	18,755	7,239	4,011	11,250	60%	30,755	11,722	3,517	36%	41,560
American Savings Bank	ASB	3,314	1,296	683	1,979	60%	5,473	2,155	647	87%	3,036
M&T Bank	MTB	20,517	5,186	6,296	11,482	56%	29,301	8,257	2,477	90%	15,562
	BOKF	3,007	1,053	621	1,674	56%	4,977	96	29	38%	4,460
Valley National	VLY	13,542	2,570	4,883	7,453	55%	17,253	3,701	1,110	198%	4,316
	ZION	7,813	2,281	1,529	3,810	49%	11,877	2,513	754	70%	6,480
Synovus	SNV	9,140	3,012	1,404	4,416	48%	12,159	619	186	93%	4,926
	HWC	3,108	569	812	1,381	45%	4,820	1,704	511	58%	3,279
	KEY	6,248	999	1,205	2,204	35%	8,852	2,530	759	20%	14,779
	CMA	3,976	576	504	1,080	27%	7,404	3,105	932	26%	7,884
	Total	**243,498**	**96,014**	**58,300**	**154,313**		**339,185**		**25,576**		
	%		**39%**	**24%**	**63%**				**8%**		

Other banks that do not disclose CRE holding details, but for which it is estimated that office and retail CRE debt exposure is significant relative to CET1 Capital include:

- **Western Alliance**
- **PacWest Bancorp**

 Sources: Bloomberg, FRED, Morgan Stanley Research, Real Capital Analytics, CoStar, JPMorgan

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